02. Normal Sliding Mode Control

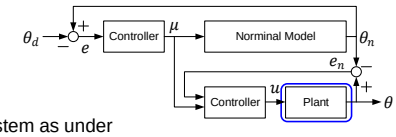
Chapter Objectives

This chapter introduce several normal sliding mode controllers design based on

- nominal model
- global sliding mode control
- linearization feedback technology
- low pass filter

§1. Sliding Mode Control Based on Nominal Model

1.System Description



Consider the system as under

$$J\ddot{\theta} + B\dot{\theta} = u - d \qquad J > 0,\ B > 0 \qquad (2.1)$$

J : moment of inertia B : damping coefficient

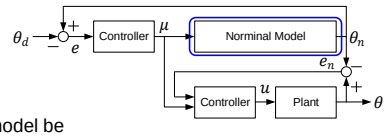
u : the control input d : disturbance

θ : rolling angle

In practice, J is the time-variant, and disturbances and uncertainties exist

§1. Sliding Mode Control Based on Nominal Model

1.System Description



Let the nominal model be

$$J_n\ddot{\theta}_n + B_n\dot{\theta}_n = \mu \qquad J_n > 0, B_n > 0 \qquad (2.2)$$

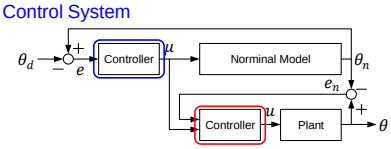
J_n: nominal moment of inertia B_n: nominal damping coefficient

μ : nominal control input θ_n: nominal rolling angle

$J\ddot{\theta} + B\dot{\theta} = u - d \quad J > 0,\ B > 0$ (2.1)

§1. Sliding Mode Control Based on Nominal Model

2.The Structure of Control System



$$J\ddot{\theta} + B\dot{\theta} = u - d \qquad J > 0,\ B > 0 \qquad (2.1)$$

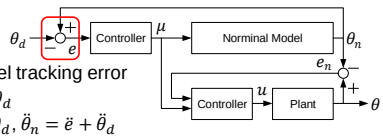
$$J_n\ddot{\theta}_n + B_n\dot{\theta}_n = \mu \qquad J_n > 0, B_n > 0 \qquad (2.2)$$

Two controllers

- Controller for nominal model $\theta_n - \theta_d \rightarrow controller \rightarrow \mu$
- Controller for actual plant $e_n, \mu \rightarrow controller \rightarrow u$

§1. Sliding Mode Control Based on Nominal Model

3.Design of Nominal Model



The nominal-model tracking error

$$e \equiv \theta_n - \theta_d$$

$$\rightarrow \dot{\theta}_n = \dot{e} + \dot{\theta}_d,\ \ddot{\theta}_n = \ddot{e} + \ddot{\theta}_d$$

and $J_n(\ddot{e} + \ddot{\theta}_d) + B_n(\dot{e} + \dot{\theta}_d)\dot{\theta}_n = \mu$

$$\ddot{e} + \ddot{\theta}_d = -\frac{B_n}{J_n}\left(\dot{e} + \dot{\theta}_d\right) + \frac{1}{J_n}\mu \qquad (2.3)$$

The controller is designed for the nominal model as

$$\mu = J_n\left(-h_1 e - h_2\dot{e} + \frac{B_n}{J_n}\dot{\theta}_d + \ddot{\theta}_d\right) \qquad (2.4)$$

$J\ddot{\theta} + B\dot{\theta} = u - d \quad J > 0, B > 0$ (2.1) $J_n\ddot{\theta}_n + B_n\dot{\theta}_n = \mu \quad J_n > 0, B_n > 0$ (2.2)

§1. Sliding Mode Control Based on Nominal Model

3.Design of Nominal Model

Proof

The tracking error

$$\ddot{e} + \ddot{\theta}_d = -\frac{B_n}{J_n}\left(\dot{e} + \dot{\theta}_d\right) + \frac{1}{J_n}\mu$$

$$\Rightarrow \ddot{e} + \left(h_2 + \frac{B_n}{J_n}\right)\dot{e} + h_1 e = -\frac{B_n}{J_n}\dot{\theta}_d + \frac{1}{J_n}\mu - \ddot{\theta}_d + h_2\dot{e} + h_1 e$$

The controller is designed for the nominal model as to make the rhs of the above equation to equal zero to get

$$\ddot{e} + \left(h_2 + \frac{B_n}{J_n}\right)\dot{e} + h_1 e = 0$$

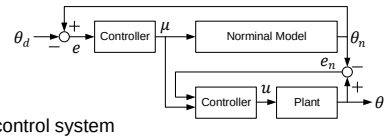
Therefore,

$$-\frac{B_n}{J_n}\dot{\theta}_d + \frac{1}{J_n}\mu - \ddot{\theta}_d + h_2\dot{e} + h_1 e = 0$$

$$\Rightarrow \mu = J_n\left(-h_1 e - h_2\dot{e} + \frac{B_n}{J_n}\dot{\theta}_d + \ddot{\theta}_d\right)$$

§1. Sliding Mode Control Based on Nominal Model

3.Design of Nominal Model



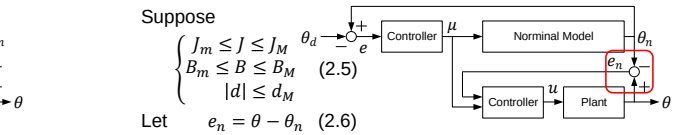
The closed-loop control system

$$\ddot{e} + \ddot{\theta}_d = -\frac{B_n}{J_n}\left(\dot{e} + \dot{\theta}_d\right) + \left(-h_1 e - h_2\dot{e} + \frac{B_n}{J_n}\dot{\theta}_d + \ddot{\theta}_d\right)$$

$$\Rightarrow \ddot{e} + \left(h_2 + \frac{B_n}{J_n}\right)\dot{e} + h_1 e = 0$$

$\ddot{e} + \ddot{\theta}_d = -\frac{B_n}{J_n}\left(\dot{e} + \dot{\theta}_d\right) + \frac{1}{J_n}\mu$ (2.3) $\quad \mu = J_n\left(-h_1 e - h_2\dot{e} + \frac{B_n}{J_n}\dot{\theta}_d + \ddot{\theta}_d\right)$ (2.4)

§1. Sliding Mode Control Based on Nominal Model



$$\ddot{e} + \left(h_2 + \frac{B_n}{J_n}\right)\dot{e} + h_1 e = 0$$

In order to make the system stable

$$p^2 + \left(h_2 + \frac{B_n}{J_n}\right)p + h_1 = 0$$

must be Hurwitz, h_1 and h_2 can be chosen as

$$h_2 + \frac{B_n}{J_n} = 2k, \qquad h_1 = k^2$$

with $k > 0$

§1. Sliding Mode Control Based on Nominal Model

4.Sliding Mode Controller Design for Actual Plant

Suppose

$$\begin{cases} J_m \le J \le J_M \\ B_m \le B \le B_M \\ |d| \le d_M \end{cases} \quad (2.5)$$

Let $e_n = \theta - \theta_n$ (2.6)

Define $s = \dot{e}_n + \lambda e_n$ (2.7)

where, $\lambda \equiv B_n/J_n > 0$ (2.8)

Denote $J_a = (J_m + J_M)/2,\ B_a = (B_m + B_M)/2$ (2.9-10)

Select the controller as

$$u = -Ks - h\,\mathrm{sgn}(s) + J_a\left(\mu/J_n - \lambda\dot{\theta}\right) + B_a\dot{\theta},\ K > 0 \quad (2.11)$$

$$h = d_M + \frac{1}{2}(J_M - J_m)\left|\frac{1}{J_n}\mu - \lambda\dot{\theta}\right| + \frac{1}{2}(B_M - B_m)|\dot{\theta}| \quad (2.12)$$

§1. Sliding Mode Control Based on Nominal Model

Proof

Let the Lyapunov function be $V = \frac{1}{2}Js^2 \ge 0$, we have

$$J\dot{s} = J\left[(\ddot{\theta} - \ddot{\theta}_n) + \lambda(\dot{\theta} - \dot{\theta}_n)\right]$$

$$= J\ddot{\theta} + B\dot{\theta} - B\dot{\theta} - \frac{J}{J_n}J_n\ddot{\theta}_n - \frac{J}{J_n}B_n\dot{\theta}_n + \frac{J}{J_n}B_n\dot{\theta}_n + J\lambda(\dot{\theta} - \dot{\theta}_n) \qquad \lambda \equiv \frac{B_n}{J_n}$$

$$= \left(J\ddot{\theta} + B\dot{\theta}\right) - \frac{J}{J_n}\left(J_n\ddot{\theta}_n + B_n\dot{\theta}_n\right) - B\dot{\theta} + \lambda J\dot{\theta}$$

$$= u - d - \frac{J}{J_n}\mu - B\dot{\theta} + \lambda J\dot{\theta}$$

$$= u - d - J\left(\frac{1}{J_n}\mu - \lambda\dot{\theta}\right) - B\dot{\theta}$$

$$= -Ks - h\,\mathrm{sgn}(s) + J_a\left(\frac{1}{J_n}\mu - \lambda\dot{\theta}\right) + B_a\dot{\theta} - d - J\left(\frac{1}{J_n}\mu - \lambda\dot{\theta}\right) - B\dot{\theta}$$

$J\ddot{\theta} + B\dot{\theta} = u - d$ (2.1) $\quad J_n\ddot{\theta}_n + B_n\dot{\theta}_n = \mu$ (2.2) $\quad e_n = \theta - \theta_n$ (2.6) $\quad s = \dot{e}_n + \lambda e_n$ (2.7) $\quad \lambda \equiv B_n/J_n$ (2.8)

$u = -Ks - h\,\mathrm{sgn}(s) + J_a\left(\mu/J_n - \lambda\dot{\theta}\right) + B_a\dot{\theta},\ K > 0$ (2.11)

$h = d_M + \frac{1}{2}(J_M - J_m)\left|\frac{1}{J_n}\mu - \lambda\dot{\theta}\right| + \frac{1}{2}(B_M - B_m)|\dot{\theta}|$ (2.12)

§1. Sliding Mode Control Based on Nominal Model

$$\dot{V} = Js\dot{s}$$

$$= s\left[-Ks - h\,\mathrm{sgn}(s) + J_a\left(\frac{1}{J_n}\mu - \lambda\dot{\theta}\right) + B_a\dot{\theta} - d - J\left(\frac{1}{J_n}\mu - \lambda\dot{\theta}\right) - B\dot{\theta}\right]$$

$$= s\left[-Ks - h\,\mathrm{sgn}(s) - d + (J_a - J)\left(\frac{1}{J_n}\mu - \lambda\dot{\theta}\right) + (B_a - B)\dot{\theta}\right]$$

$$= -Ks^2 - h|s| + s\left[-d + (J_a - J)\left(\frac{1}{J_n}\mu - \lambda\dot{\theta}\right) + (B_a - B)\dot{\theta}\right]$$

$$\le -Ks^2 - h|s| + |s|\left(|d| + |J_a - J|\left|\frac{1}{J_n}\mu - \lambda\dot{\theta}\right| + |B_a - B||\dot{\theta}|\right)$$

with $J_a = \frac{J_m + J_M}{2} \rightarrow |J_a - J| \le \frac{J_M - J_m}{2}$, $\quad B_a = \frac{B_m + B_M}{2} \rightarrow |B_a - B| \le \frac{B_M - B_m}{2}$

$$\Rightarrow h = d_M + \frac{J_M - J_m}{2}\left|\frac{1}{J_n}\mu - \lambda\dot{\theta}\right| + \frac{B_M - B_m}{2}|\dot{\theta}| \ge |d| + |J_a - J|\left|\frac{1}{J_n}\mu - \lambda\dot{\theta}\right| + |B_a - B||\dot{\theta}|$$

or $\quad -h|s| + |s|\left(|d| + |J_a - J|\left|\frac{1}{J_n}\mu - \lambda\dot{\theta}\right| + |B_a - B||\dot{\theta}|\right) \le 0$

$$\Rightarrow \dot{V} \le -Ks^2 - h|s| + |s|\left(|d| + |J_a - J|\left|\frac{1}{J_n}\mu - \lambda\dot{\theta}\right| + |B_a - B||\dot{\theta}|\right)$$

$$\le 0$$

§1. Sliding Mode Control Based on Nominal Model

Summarize

The system

$J\ddot{\theta} + B\dot{\theta} = u - d$

The nominal model

$J_n\ddot{\theta}_n + B_n\dot{\theta}_n = \mu$

with, $J_n = k^2$, $B_n = 2k - B_n/J_n$

The controller for the nominal model

$\mu = J_n[-h_1 e - h_2\dot{e} + (B_n/J_n)\dot{\theta}_d + \ddot{\theta}_d]$

with, $h_1 = k^2$, $h_2 = 2k - B_n/J_n$, $e = \theta_n - \theta_d$

The controller for the plant

$u = -Ks - h\mathrm{sgn}(s) + J_a(\mu/J_n - \lambda\dot{\theta}) + B_a\dot{\theta}$, $K > 0$

with, $e_n = \theta - \theta_n$

$s = \dot{e}_n + \lambda e_n$

$h = d_M + \frac{1}{2}(J_M - J_m)\left|\frac{1}{J_n}\mu - \lambda\dot{\theta}\right| + \frac{1}{2}(B_M - B_m)|\dot{\theta}|$

$\lambda = B_n/J_n$, $J_a = (J_m + J_M)/2$, $B_a = (B_m + B_M)/2$, $J_m \le J \le J_M$, $B_m \le B \le B_M$, $|d| \le d_M$

§1. Sliding Mode Control Based on Nominal Model

5. Simulation (Ex_02.01.m)

Consider the system as follows

$J\ddot{\theta} + B\dot{\theta} = u - d$

where, $B = 10 + 3\sin 2\pi t$, $J = 3 + 0.5\sin 2\pi t$, $d = 10\sin t$

The desired trajectory

$\theta_d(t) = \sin t$

and the initial state vector

$[\theta(0), \dot{\theta}(0)] = [0.5, 0]$

$J_m \le J \le J_M$, $B_m \le B \le B_M$, $|d| \le d_M$ (2.5)

§1. Sliding Mode Control Based on Nominal Model

5. Simulation

System $\begin{bmatrix}\dot{\theta}\\ \ddot{\theta}\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & -\frac{B}{J}\end{bmatrix}\begin{bmatrix}\theta\\ \dot{\theta}\end{bmatrix} + \begin{bmatrix}0\\ \frac{1}{J}\end{bmatrix}u + \begin{bmatrix}0\\ -\frac{1}{J}\end{bmatrix}d$

$\theta_d(t) = \sin t$, $[\theta(0), \dot{\theta}(0)] = [0.5, 0]$

$B = 10 + 3\sin 2\pi t \rightarrow B_m = 7, B_M = 17, B_n = 10$

$J = 3 + 0.5\sin 2\pi t \rightarrow J_m = 2.5, J_M = 3.5, J_n = 3$

$d = 10\sin t \rightarrow d_M = 10$

Controller $k = 1 \rightarrow h_1 = k^2 = 1, h_2 = 2k - \frac{B_n}{J_n} = 1 - \frac{10}{3} = -\frac{7}{3}$

$\lambda = \frac{B_n}{J_n} = \frac{10}{3}$

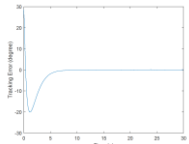
$K = 10$

$J_m \le J \le J_M$, $B_m \le B \le B_M$, $|d| \le d_M$ (2.5)

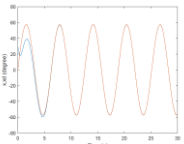
$h_1 = k^2$, $h_2 = B_n/J_n = 2k$, $\lambda \equiv B_n/J_n > 0$ (2.8)

§1. Sliding Mode Control Based on Nominal Model

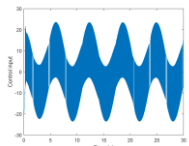
Results with using sign function



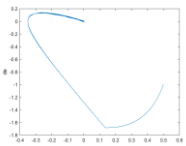
Tracking error



Position tracking



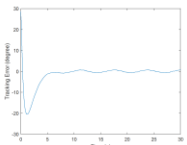
Control input



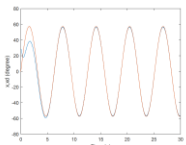
Phase trajectory

§1. Sliding Mode Control Based on Nominal Model

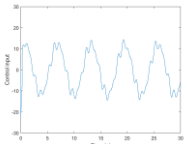
Results with using saturation function



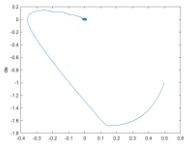
Tracking error



Position tracking



Control input



Phase trajectory

§2. Global Sliding Mode Control for an Uncertain System

1. System Description

Consider the system as follows

$J\ddot{\theta} = u(t) - d(t)$ (2.13)

We can get

$\ddot{\theta} = b(u(t) - d(t))$

where, J : the moment of inertia, $J_{min} \le J \le J_{max}$ (2.14)

$b = \frac{1}{J} > 0$

$d(t)$: the disturbance, $|d(t)| < D$ (2.15)

§2. Global Sliding Mode Control for an Uncertain System

2. Global Sliding Mode Design

Let the desired trajectory be θ_d and the tracking error be

$$e = \theta - \theta_d \tag{2.16}$$

The global dynamic sliding mode variable is designed as

$$s = \dot{e} + ce - f(t) \tag{2.17}$$

$$b = 1/J > 0$$

where, $c > 0$

$f(t)$: function to be designed to satisfy

(1) $f(0) = \dot{e}_0 + ce_0$

(2) $f(t) \to 0$ as $t \to \infty$

(3) $f(t)$ is derivable

$$f(t) = f(0)e^{-kt} \tag{2.18}$$

$J\ddot{\theta} = u(t) - d(t)$ (2.13)

$\ddot{\theta} = b(J\ddot{\theta} = u(t) - d(t))$, $J_{min} \le J \le J_{max}$ (2.14), $b = 1/J > 0$, $|d(t)| < D$ (2.15)

§2. Global Sliding Mode Control for an Uncertain System

3. Sliding Mode Controller Design

The global sliding mode controller is designed as

$$u = -\hat{J}(c\dot{\theta} - \dot{f}) + \hat{J}(\ddot{\theta}_d - c\dot{\theta}_d) - (\Delta J|c\dot{\theta} - \dot{f}| + D + \Delta J|\ddot{\theta}_d - c\dot{\theta}_d|)\text{sgn}(s) \tag{2.19}$$

where, $\hat{J} = \dfrac{J_{max} + J_{min}}{2}$, $\Delta J = \dfrac{J_{max} - J_{min}}{2}$ (2.20)

Let the Lyapunov function be $V = \dfrac{1}{2}s^2$

(2.17) $\to \dot{s} = \ddot{e} + c\dot{e} - \dot{f}$

$$= \ddot{\theta} - \ddot{\theta}_d + c(\dot{\theta} - \dot{\theta}_d) - \dot{f}$$

$$= bu - bd + (c\dot{\theta} - \dot{f}) - (\ddot{\theta}_d + c\dot{\theta}_d)$$

$$= b\left[b^{-1}(c\dot{\theta} - \dot{f}) - b^{-1}(\ddot{\theta}_d + c\dot{\theta}_d) + u - d\right]$$

$s = \dot{e} + ce - f(t)$ (2.17)

§2. Global Sliding Mode Control for an Uncertain System

3. Sliding Mode Controller Design

(2.19) $\to b^{-1}\dot{s} = b^{-1}(c\dot{\theta} - \dot{f}) - b^{-1}(\ddot{\theta}_d + c\dot{\theta}_d)$

$$-\hat{J}(c\dot{\theta} - \dot{f}) + \hat{J}(\ddot{\theta}_d + c\dot{\theta}_d)$$

$$-(\Delta J|c\dot{\theta} - \dot{f}| + D + \Delta J|\ddot{\theta}_d + c\dot{\theta}_d|)\,\text{sgn}(s) - d$$

$$= (b^{-1} - \hat{J})(c\dot{\theta} - \dot{f}) - \Delta J|c\dot{\theta} - \dot{f}|\,\text{sgn}(s)$$

$$-(b^{-1} - \hat{J})(\ddot{\theta}_d + c\dot{\theta}_d) - \Delta J|\ddot{\theta}_d + c\dot{\theta}_d|\,\text{sgn}(s)$$

$$-d - D\text{sgn}(s)$$

Therefore

$$b^{-1}\dot{V} = b^{-1}s\dot{s}$$

$$= (b^{-1} - \hat{J})(c\dot{\theta} - \dot{f})s - \Delta J|c\dot{\theta} - \dot{f}||s|$$

$$-(b^{-1} - \hat{J})(\ddot{\theta}_d + c\dot{\theta}_d)s - \Delta J|\ddot{\theta}_d + c\dot{\theta}_d||s|$$

$$-ds - D|s|$$

$u = -\hat{J}(c\dot{\theta} - \dot{f}) + \hat{J}(\ddot{\theta}_d - c\dot{\theta}_d) - (\Delta J|c\dot{\theta} - \dot{f}| + D + \Delta J|\ddot{\theta}_d - c\dot{\theta}_d|)\text{sgn}(s)$ (2.19)

§2. Global Sliding Mode Control for an Uncertain System

3. Sliding Mode Controller Design

(2.20) $\to b^{-1} - \hat{J} = J - \dfrac{J_{max} + J_{min}}{2} \le \dfrac{J_{max} - J_{min}}{2} = \Delta J > 0$

Therefore

$$b^{-1}\dot{V} < -ds - D|s| < 0 \quad \text{or} \quad \dot{V} < 0$$

In order to reduce the chattering phenomenon, the saturated function can be used, i.e.

$$\text{sat}\left(\frac{s}{\varphi}\right) = \begin{cases} 1 & \frac{s}{\varphi} > 1 \\ \frac{s}{\varphi} & \left|\frac{s}{\varphi}\right| \le 1 \\ -2 & \frac{s}{\varphi} < -1 \end{cases} \tag{2.22}$$

$\hat{J} = \frac{J_{max}+J_{min}}{2}$, $\Delta J = \frac{J_{max}-J_{min}}{2}$ (2.20)

§2. Global Sliding Mode Control for an Uncertain System

Summarize

The system

$J\ddot{\theta} = u - d$ or $\ddot{\theta} = b(u - d)$

with, $b = 1/J$

The controller

$u = -\hat{J}(c\dot{\theta} - \dot{f}) + \hat{J}(\ddot{\theta}_d - c\dot{\theta}_d) - (\Delta J|c\dot{\theta} - \dot{f}| + D + \Delta J|\ddot{\theta}_d - c\dot{\theta}_d|)\text{sgn}(s)$

with, $\hat{J} = \dfrac{J_{max} + J_{min}}{2}$

$e = \theta - \theta_d$

$s = \dot{e} + ce - f(t)$

$f(t) = f(0)e^{-kt}$

$\Delta J = \dfrac{J_{max} - J_{min}}{2}$

$|d(t)| < D$

§2. Global Sliding Mode Control for an Uncertain System

4. Simulation Example (Ex_02.02.m)

Let the plant be

$$J\ddot{\theta} = u(t) - d(t)$$

where, $J = 1.0 + 0.2sint$, $d = 0.1\sin 2\pi t$

Reference trajectory $\theta_d = sint$

Initial values $[\theta(0), \dot{\theta}(0)] = [\pi/6, 0]$

§2. Global Sliding Mode Control for an Uncertain System

4. Simulation Example

The system

$$\begin{bmatrix}\dot{\theta}\\ \ddot{\theta}\end{bmatrix}=\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}\theta\\ \dot{\theta}\end{bmatrix}+\begin{bmatrix}0\\ \frac{1}{J}\end{bmatrix}u+\begin{bmatrix}0\\ -\frac{1}{J}\end{bmatrix}d$$

Here, $\theta_d = sint$

$d = 0.1\sin 2\pi t \quad \rightarrow D = 0.1$

$J = 1.0 + 0.2sint \quad \rightarrow J_m = 0.8, J_M = 1.2$

$\hat{J} = \frac{J_{max}+J_{min}}{2} = 1.0, \quad \Delta J = \frac{J_{max}-J_{min}}{2} = 0.2$

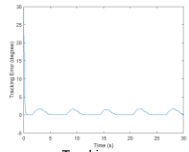
Controller $c = 10$, $f(t) = s(0)e^{-130t}$, $\varphi = 0.05$

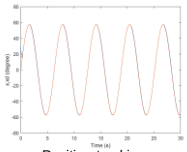
$$u = -\hat{J}(c\dot{\theta} - \hat{f}) + \hat{J}(\ddot{\theta}_d - c\dot{\theta}_d) - (\Delta J|c\dot{\theta} - \hat{f}| + D + \Delta J|\ddot{\theta}_d - c\dot{\theta}_d|)\mathrm{sgn}(s) \quad (2.19)$$

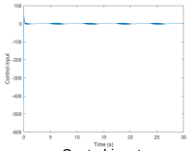
$$\hat{J} = \frac{J_{max}+J_{min}}{2}, \quad \Delta J = \frac{J_{max}-J_{min}}{2} \quad (2.20)$$

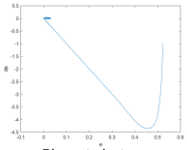
§2. Global Sliding Mode Control for an Uncertain System

Results with using sign function


Tracking error

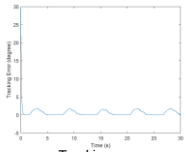

Position tracking

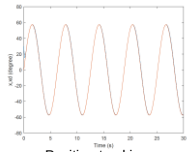

Control input

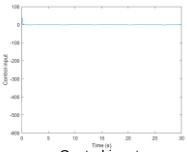

Phase trajectory

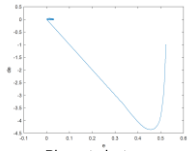
§2. Global Sliding Mode Control for an Uncertain System

Results with using saturation function


Tracking error


Position tracking


Control input


Phase trajectory

§3. Sliding Mode Control Based on Linearization Feedback Control

1. Linearization Feedback Control

Consider the following second-order SISO nonlinear system

$$\ddot{x} = f(x,t) + g(x,t)u \quad (2.23)$$

where f and g are known nonlinear functions

Let x_d : the desired trajectory

e : the tracking error, $e = x_d - x$

Based on linearization feedback technique, sliding mode controller is designed as

$$u = \frac{v - f(x,t)}{g(x,t)} \quad (2.24)$$

where v is the auxiliary controller

§3. Sliding Mode Control Based on Linearization Feedback Control

1. Linearization Feedback Control

The auxiliary controller v will be designed in the following

From (2.24) and (2.23), we have

$$\ddot{x} = v \quad (2.25)$$

v is designed as

$$v = \ddot{x}_d + k_1 e + k_2 \dot{e} \quad (2.26)$$

where, k_1 and k_2 are positive constants

From (2.26) and (2.25), we get

$$\ddot{e} + k_1 e + k_2 \dot{e} = 0 \quad (2.27)$$

Therefore, $e_1 \rightarrow 0$ and $e_2 \rightarrow 0$ as $t \rightarrow \infty$

$\ddot{x} = f(x,t) + g(x,t)u$ (2.23) $u = \frac{v - f(x,t)}{g(x,t)}$ (2.24)

§3. Sliding Mode Control Based on Linearization Feedback Control

Summarize

The system

$\ddot{x} = f(x,t) + g(x,t)u$

with f and g are known nonlinear functions

The controller

$u = \frac{v - f(x,t)}{g(x,t)}$

with, $e = x_d - x$

$v = \ddot{x}_d + k_1 e + k_2 \dot{e}$

§3. Sliding Mode Control Based on Linearization Feedback Control

2.Simulation Example (Ex_02.03.m)

The dynamic equation of inverted pendulum

$$\begin{cases}\dot{x}_1 = x_2 \\ \dot{x}_2 = f(\boldsymbol{x}) + g(\boldsymbol{x})u\end{cases}$$

$$f(\boldsymbol{x}) = \frac{g\sin x_1 - mlx_2^2\cos x_1 \sin x_1/(m_c+m)}{l[4/3 - m\cos^2 x_1/(m_c+m)]}$$

$$g(\boldsymbol{x}) = \frac{\cos x_1/(m_c+m)}{l[4/3 - m\cos^2 x_1/(m_c+m)]}$$

x_1 : the oscillation angle θ x_2 : the oscillation rate $\dot{\theta}$

g : gravity acceleration, $g = 9.81m/s^2$

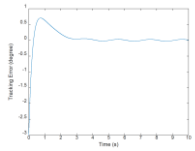
m_c : vehicle mass, $m_c = 1kg$ m : pendulum mass, $m = 0.1kg$

$2l$: pendulum length, $2l = 1m$ u : the control input

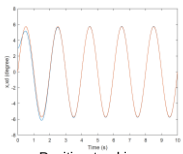
Desired trajectory $x_d = 0.1sin\pi t$; initial state $[\pi/60 \quad 0]$; $k_1 = k_2 = 5$

§3. Sliding Mode Control Based on Linearization Feedback Control

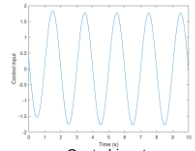
Results with linearization feedback controller



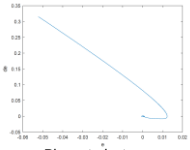
Tracking error



Position tracking



Control input



Phase trajectory

§3. Sliding Mode Control Based on Linearization Feedback Control

3.Sliding Mode Control Based on Linearization Feedback

Consider the following second-order SISO uncertain nonlinear system

$$\ddot{x} = f(x,t) + g(x,t)u + d(t) \quad (2.28)$$

where f and g are known nonlinear functions, d is the uncertainty

Let the desired trajectory be $\boldsymbol{x}_d$, the tracking error

$$\boldsymbol{e} = \boldsymbol{x} - \boldsymbol{x}_d = [e \quad \dot{e}]^T \quad (2.29)$$

Sliding variable is selected as

$$s(x,t) = \boldsymbol{ce}, \quad \boldsymbol{c} = [c \quad 1], \quad \boldsymbol{e} = [e \quad \dot{e}]^T \quad (2.30)$$

Based on linearization feedback technique, the sliding mode controller is designed as

$$u = \frac{v - f(x,t)}{g(x,t)} \quad (2.31)$$

$$v = \ddot{x}_d - c\dot{e} - \eta\,\mathrm{sgn}(s),\ \eta > D \quad (2.32)$$

§3. Sliding Mode Control Based on Linearization Feedback Control

3.Sliding Mode Control Based on Linearization Feedback

Proof

Select the Lyapunov function V $v = f(x,t) + g(x,t)u$

$$V = \frac{1}{2}s^2$$

$$\Rightarrow \dot{V} = s\dot{s}$$
$$= s(\ddot{e} + c\dot{e})$$
$$= s(\ddot{x} - \ddot{x}_d + c\dot{e})$$
$$= s[f(x,t) + g(x,t)u + d - \ddot{x}_d + c\dot{e}]$$
$$= s(v + d - \ddot{x}_d + c\dot{e})$$
$$= s(\ddot{x}_d - c\dot{e} - \eta\,\mathrm{sgn}(s) + d - \ddot{x}_d + c\dot{e})$$
$$= s(-\eta\,\mathrm{sgn}(s) + d)$$
$$= -\eta|s| + ds$$
$$\leq 0$$

$\ddot{x} = f(x,t) + g(x,t)u + d(t)$ (2.28) $e = x - x_d$ (2.29) $s = ce + \dot{e}$ (2.30)

$u = \frac{v - f(x,t)}{g(x,t)}$ (2.31) $v = \ddot{x}_d - c\dot{e} - \eta\,\mathrm{sgn}(s),\ \eta > D$ (2.32)

§3. Sliding Mode Control Based on Linearization Feedback Control

Summarize

The system

$\ddot{x} = f(x,t) + g(x,t)u + d(t)$

with f and g are known nonlinear functions, d is the uncertainty

The controller

$u = \frac{v - f(x,t)}{g(x,t)}$

with, $e = x - x_d$

$s = ce + \dot{e}$

$v = \ddot{x}_d - c\dot{e} - \eta\,\mathrm{sgn}(s)$, $\eta > D$

Relay function

$\mathrm{relay}(s) = \frac{s}{|s| + \delta}$

$\delta = \delta_0 + \delta_1|e|$

§3. Sliding Mode Control Based on Linearization Feedback Control

4.Simulation Example (Ex_02.04.m)

The dynamic equation of inverted pendulum

$$\begin{cases}\dot{x}_1 = x_2 \\ \dot{x}_2 = f(\boldsymbol{x}) + g(\boldsymbol{x})u + d(t)\end{cases}$$

$$f(\boldsymbol{x}) = \frac{g\sin x_1 - mlx_2^2\cos x_1 \sin x_1/(m_c+m)}{l[4/3 - m\cos^2 x_1/(m_c+m)]}$$

$$g(\boldsymbol{x}) = \frac{\cos x_1/(m_c+m)}{l[4/3 - m\cos^2 x_1/(m_c+m)]}$$

x_1 : the oscillation angle θ x_2 : the oscillation rate $\dot{\theta}$

g : gravity acceleration, $g = 9.81m/s^2$

m_c : vehicle mass, $m_c = 1kg$ m : pendulum mass, $m = 0.1kg$

$2l$: pendulum length, $2l = 1m$ u : the control input

Desired trajectory $x_d = 0.1sin\pi t$; initial state $[\pi/60 \quad 0]$

§3. Sliding Mode Control Based on Linearization Feedback Control

4. Simulation Example

The dynamic equation of inverted pendulum

$$\begin{cases}\dot{x}_1 = x_2 \\ \dot{x}_2 = f(\boldsymbol{x}) + g(\boldsymbol{x})u + d(t)\end{cases}$$

$$f(\boldsymbol{x}) = \frac{g\sin x_1 - mlx_2^2\cos x_1 \sin x_1/(m_c+m)}{l[4/3 - m\cos^2 x_1/(m_c+m)]}$$

$$g(\boldsymbol{x}) = \frac{\cos x_1/(m_c+m)}{l[4/3 - m\cos^2 x_1/(m_c+m)]}$$

Using sign function

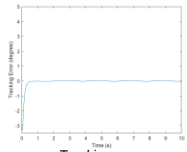
Using relay function function

$$relay(s) = \frac{s}{|s| + \delta}$$

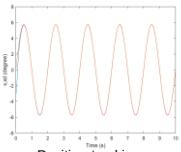
with $\delta_0 = 0.03$, $\delta_1 = 5$, $\delta = \delta_0 + \delta_1|\boldsymbol{e}|$

§3. Sliding Mode Control Based on Linearization Feedback Control

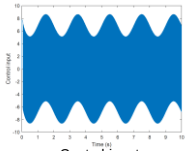
Results with using sign function



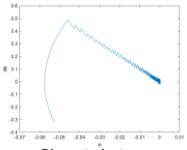
Tracking error



Position tracking



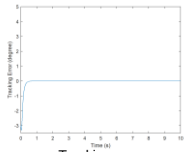
Control input



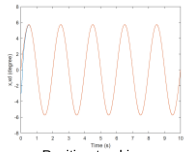
Phase trajectory

§3. Sliding Mode Control Based on Linearization Feedback Control

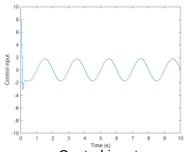
Results with using delayed function



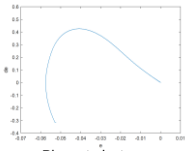
Tracking error



Position tracking



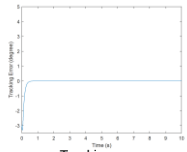
Control input



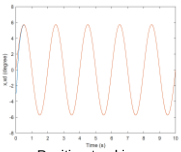
Phase trajectory

§3. Sliding Mode Control Based on Linearization Feedback Control

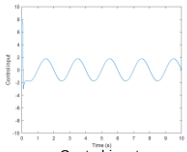
Results with using saturation function



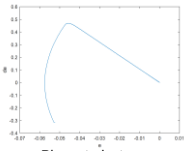
Tracking error



Position tracking



Control input



Phase trajectory

§4. Input-Output Feedback Linearization Control

1. System Description

Consider the following system

$$\begin{cases}\dot{x}_1 = \sin x_2 + (x_2+1)x_3 \\ \dot{x}_2 = x_1^5 + x_3 \\ \dot{x}_3 = x_1^2 + u \\ y = x_1\end{cases} \quad (2.33)$$

Problem statement: to make the output y track the desired trajectory y_d

From Eq. (2.33), the output y is connected with u indirectly through $x \rightarrow$ It is difficult to design the controller

§4. Input-Output Feedback Linearization Control

2. Controller Design

In order to obtain the relation of y and u, differentiate y until u appears

$$\begin{aligned}\dot{y} &= \dot{x}_1 \\ &= \sin x_2 + (x_2+1)x_3 \end{aligned} \quad (2.34)$$

$$\begin{aligned}\ddot{y} &= \ddot{x}_1 \\ &= \dot{x}_2\cos x_2 + \dot{x}_2 x_3 + (x_2+1)\dot{x}_3 \\ &= (x_1^5 + x_3)\cos x_2 + (x_1^5 + x_3)x_3 + (x_2+1)(x_1^2+u) \\ &= (x_1^5 + x_3)(\cos x_2 + x_3) + (x_2+1)x_1^2 + (x_2+1)u \end{aligned} \quad (2.35)$$

Let $f(x) = (x_1^5 + x_3)(\cos x_2 + x_3) + (x_2+1)x_1^2$

therefore,

$$\ddot{y} = (x_2+1)u + f(x) \quad (2.36)$$

$\dot{x}_1 = \sin x_2 + (x_2+1)x_3, \quad \dot{x}_1 = x_1^5 + x_3, \quad \dot{x}_3 = x_1^2 + u, \quad y = x_1$ (2.33)

§4. Input-Output Feedback Linearization Control

2.Controller Design

Eq.(2.36) denotes the visible relation of the output y and the input u

Select $u = \frac{1}{x_2 + 1}(v - f)$ (2.37)

v : the auxiliary controller and will be designed in the following

From (2.37) and (2.36)

$$\ddot{y} = v \quad (2.38)$$

Denote $e = y_d - y$, design v with PD and feedforward froms

$$v = \ddot{y}_d + k_1 e + k_2 \dot{e}, \quad k_1, k_2 > 0 \quad (2.39)$$

From (2.39) and (2.38)

$$\ddot{e} + k_2 \dot{e} + k_1 e = 0 \quad (2.40)$$

Therefore $e_1 \to 0$ and $e_2 \to 0$ as $t \to \infty$

$\ddot{y} = (x_2 + 1)u + f(x)$ (2.36)

§4. Input-Output Feedback Linearization Control

3.Simulation Example (Ex_02.05.m)

$$\begin{cases} \dot{x}_1 = \sin x_2 + (x_2 + 1)x_3 \\ \dot{x}_2 = x_1^5 + x_3 \\ \dot{x}_3 = x_1^2 + u \\ y = x_1 \end{cases}$$

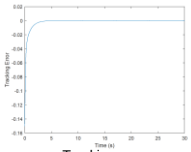
The desired trajectory is $y_d = \sin t$. Let $k_1 = k_2 = 10$

The controller is

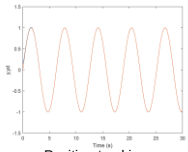
$$u = \frac{1}{x_2 + 1}(v - f)$$
$$e = y_d - y$$
$$v = \ddot{y}_d + k_1 e + k_2 \dot{e}$$
$$f(x) = (x_1^5 + x_3)(\cos x_2 + x_3) + (x_2 + 1)x_1^2$$

§4. Input-Output Feedback Linearization Control

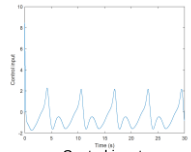
Results



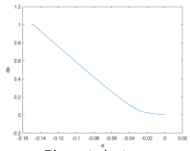
Tracking error



Position tracking



Control input



Phase trajectory

§5. Sliding Mode Control Based on Input-Output Feedback Linearization

Sliding mode robust item is introduced into the controller to overcome the faults in the usual input-output feedback linearization sliding mode control. Then, the input-output feedback linearization sliding mode control is designed

1.System Description

Consider the following uncertain system

$$\begin{cases} \dot{x}_1 = \sin x_2 + (x_2 + 1)x_3 + d_1 \\ \dot{x}_2 = x_1^5 + x_3 + d_2 \\ \dot{x}_3 = x_1^2 + u + d_3 \\ y = x_1 \end{cases} \quad (2.41)$$

d_1, d_2, d_3: the uncertainties in the three sub-systems

Problem statement: to make the output y track the desired trajectory y_d. From Eq. (2.41), the output y is connected with u indirectly through $x \to$ It is difficult to design the controller

§5. Sliding Mode Control Based on Input-Output Feedback Linearization

2.Controller Design

In order to obtain the relation of y and u, differentiate y until u appears

$$\dot{y} = \dot{x}_1 = \sin x_2 + (x_2 + 1)x_3 + d_1 \quad (2.42)$$

$$\begin{aligned} \ddot{y} &= \ddot{x}_1 \\ &= \dot{x}_2 \cos x_2 + \dot{x}_2 x_3 + (x_2 + 1)\dot{x}_3 + \dot{d}_1 \\ &= (x_1^5 + x_3 + d_2)\cos x_2 + (x_1^5 + x_3 + d_2)x_3 \\ &\quad + (x_2 + 1)(x_1^2 + u + d_3) + \dot{d}_1 \\ &= (x_1^5 + x_3)(\cos x_2 + x_3) + (x_2 + 1)x_1^2 + (x_2 + 1)u + d \end{aligned} \quad (2.43)$$

where, $d = d_2 \cos x_2 + d_2 x_3 + (x_2 + 1)d_3 + d_1$, $|d| \le D$

Let $f(x) = (x_1^5 + x_3)(\cos x_2 + x_3) + (x_2 + 1)x_1^2$

$$\Rightarrow \ddot{y} = (x_2 + 1)u + f(x) + d \quad (2.44)$$

$\dot{x}_1 = \sin x_2 + (x_2 + 1)x_3 + d_1, \quad \dot{x}_2 = x_1^5 + x_3 + d_2, \quad \dot{x}_3 = x_1^2 + u + d_3, \quad y = x_1$ (2.41)

§5. Sliding Mode Control Based on Input-Output Feedback Linearization

2.Controller Design

Denote $e = y_d - y$, and the sliding variable is selected as

$$s(x,t) = \boldsymbol{ce}, \quad \boldsymbol{c} = [c \;\; 1], \quad \boldsymbol{e} = [e \;\; \dot{e}]^T \quad (2.45)$$

(2.44) denotes the visible relation of the output y and the input u

Select $u = \frac{1}{x_2 + 1}(v - f + \eta \,\mathrm{sgn}(s)), \qquad \eta \ge D \quad (2.46)$

v : the auxiliary controller and will be designed in the following

$\ddot{y} = (x_2 + 1)u + f(x) + d$ (2.44)

§5. Sliding Mode Control Based on Input-Output Feedback Linearization

2.Controller Design

Select the Lyapunov function $V = \frac{1}{2}s^2$

$$\begin{aligned}\dot{V} &= s\dot{s}\\ &= s(\ddot{e} + c\dot{e})\\ &= s(\ddot{y}_d - \ddot{y} + c\dot{e})\\ &= s[\ddot{y}_d - (x_2+1)u - f(x) - d + c\dot{e}]\\ &= s\{\ddot{y}_d - [v - f(x) + \eta\,\mathrm{sgn}(s)] - f(x) - d + c\dot{e}\}\\ &= s[\ddot{y}_d - v - \eta\,\mathrm{sgn}(s) - d + c\dot{e}] \qquad (2.47)\end{aligned}$$

Select v be

$$v = \ddot{y}_d + c\dot{e} \qquad (2.48)$$

$$\Rightarrow \dot{V} = s(-\eta\,\mathrm{sgn}(s) - d) = ds - \eta|s| \le (D - \eta)|s| \le 0$$

$$u = \frac{1}{x_2+1}(v - f + \eta\,\mathrm{sgn}(s)),\quad \eta \ge D \qquad (2.46)$$

§5. Sliding Mode Control Based on Input-Output Feedback Linearization

3.Simulation Example (Ex_02.06.m)

$$\begin{cases}\dot{x}_1 = \sin x_2 + (x_2+1)x_3 + d_1\\ \dot{x}_2 = x_1^5 + x_3 + d_2\\ \dot{x}_3 = x_1^2 + u + d_3\\ y = x_1\end{cases}$$

$d_1 = d_2 = d_3 = sint$

The desired trajectory is $y_d = \sin t$. Let $c = 10$, $\eta = 3.0$

The controller is

$$u = \frac{1}{x_2+1}(v - f + \eta\,\mathrm{sgn}(s))$$

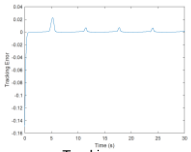
$$e = y_d - y$$

$$v = \ddot{y}_d + c\dot{e}$$

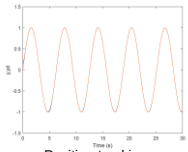
$$f(x) = (x_1^5 + x_3)(\cos x_2 + x_3) + (x_2+1)x_1^2$$

§5. Sliding Mode Control Based on Input-Output Feedback Linearization

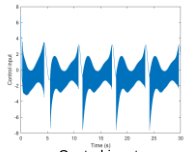
Results



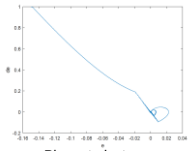
Tracking error



Position tracking



Control input



Phase trajectory

§6. Sliding Mode Control Based on Low Pass Filter

1.System Description

Consider the following second servo system

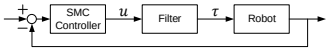
$$J\ddot{\theta} = \tau - d(t) \qquad (2.49)$$

where J is the initial moment, τ is the control input, $d(t)$ is the disturbance

§6. Sliding Mode Control Based on Low Pass Filter

2.Sliding Mode Controller Design

Sliding mode control system with low pass filter



$u(t)$: the virtual control input, $\tau(t)$: the practical control input

To decrease control chattering, low pass filter is designed as

$$Q(s) = \frac{\lambda}{s+\lambda}, \qquad \lambda > 0 \qquad (2.50)$$

From the figure

$$\dot{\tau} + \lambda\tau = \lambda u \qquad (2.51)$$

$$J\ddot{\theta} + d(t) = \tau$$

$$\Rightarrow J\dddot{\theta} + \dot{d} + \lambda(J\ddot{\theta} + d) = \lambda u$$

and $J\dddot{\theta} = \lambda u - \dot{d} - \lambda(J\ddot{\theta} + d)$

§6. Sliding Mode Control Based on Low Pass Filter

2.Sliding Mode Controller Design

Assume ideal position signal is $\theta_d(t)$, the tracking error is

$$e(t) = \theta(t) - \theta_d(t) \qquad (2.52)$$

Sliding function is designed as

$$s(t) = \ddot{e} + \lambda_1\dot{e} + \lambda_2 e, \quad \lambda_1,\lambda_2 > 0 \qquad (2.53)$$

Then $J\dot{s} = J(\dddot{e} + \lambda_1\ddot{e} + \lambda_2\dot{e})$

$$= J\dddot{\theta} + J(-\dddot{\theta}_d + \lambda_1\ddot{e} + \lambda_2\dot{e})$$

$$= \lambda u - \dot{d} - \lambda(J\ddot{\theta} + d) + J(-\dddot{\theta}_d + \lambda_1\ddot{e} + \lambda_2\dot{e})$$

Define Lyapunov function $V = \frac{1}{2}s^2$, then

$$\dot{V} = Js\dot{s}$$

$$= s[\lambda u - \dot{d} - \lambda(J\ddot{\theta} + d) + J(-\dddot{\theta}_d + \lambda_1\ddot{e} + \lambda_2\dot{e})]$$

$$= s[\lambda u - \dot{d} - \lambda d - \lambda J\ddot{\theta} + J(-\dddot{\theta}_d + \lambda_1\ddot{e} + \lambda_2\dot{e})]$$

§6. Sliding Mode Control Based on Low Pass Filter

2. Sliding Mode Controller Design

Define Lyapunov function $V = \frac{1}{2}s^2$, then

$$\begin{aligned}\dot{V} &= Js\dot{s}\\ &= s[\lambda u - \dot{d} - \lambda(J\ddot{\theta} + d) + J(-\dddot{\theta}_d + \lambda_1\ddot{e} + \lambda_2\dot{e})]\\ &= s[\lambda u - \dot{d} - \lambda d - \lambda J\ddot{\theta} + J(-\dddot{\theta}_d + \lambda_1\ddot{e} + \lambda_2\dot{e})]\end{aligned}$$

Sliding mode controller is designed as

$$u = -\frac{1}{\lambda}[-\lambda J\ddot{\theta} + J(-\dddot{\theta}_d + \lambda_1\ddot{e} + \lambda_2\dot{e}) + \eta\,\mathrm{sgn}(s)] \quad (2.55)$$

where, $\eta > |\dot{d} + \lambda d|$

Then $\dot{V} = s(-\eta\,\mathrm{sgn}(s) - \dot{d} - \lambda d)$

$$\begin{aligned}&= -s(\dot{d} + \lambda d) - \eta s\,\mathrm{sgn}(s)\\ &= -s(\dot{d} + \lambda d) - \eta|s|\\ &< 0\end{aligned}$$

§6. Sliding Mode Control Based on Low Pass Filter

Summarize

The system $J\ddot{\theta} = \tau - d(t)$



with τ is the control input, $d(t)$ is the disturbance

The controller

$$u = -\frac{1}{\lambda}[-\lambda J\ddot{\theta} + J(-\dddot{\theta}_d + \lambda_1\ddot{e} + \lambda_2\dot{e}) + \eta\,\mathrm{sgn}(s)]$$

with, $e = \theta - \theta_d$

$s = \ddot{e} + \lambda_1\dot{e} + \lambda_2 e$, $\quad \lambda_1, \lambda_2 > 0$

$\dot{\tau} + \lambda\tau = \lambda u$ $\quad \lambda > 0$

Relay function

$$\mathrm{relay}(s) = \frac{s}{|s| + \delta}$$

$$\delta = \delta_0 + \delta_1|e|$$

§6. Sliding Mode Control Based on Low Pass Filter

3. Simulation Example (Ex_02.07.m)

The plant $J\ddot{\theta} = \tau - d(t)$

$J = 10$

$d(t) = 0.3\sin t$

The ideal position signal $r = \sin t$

The initial plant state $[0.5 \quad 0]^T$

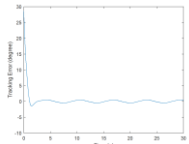
Using the controller (2.55) with low pass filter (2.50)

Rewrite the system

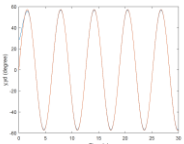
$$\begin{bmatrix}\dot{\theta}\\ \ddot{\theta}\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}\theta\\ \dot{\theta}\end{bmatrix} + \begin{bmatrix}0\\ \frac{1}{J}\end{bmatrix}\tau + \begin{bmatrix}0\\ -\frac{1}{J}\end{bmatrix}d$$

§6. Sliding Mode Control Based on Low Pass Filter

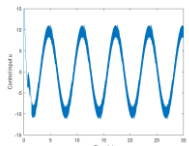
Results without low pass filter $\lambda = 25, \lambda_1 = 04, \lambda_2 = 12, \eta = 20$



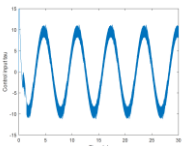
Tracking error



Position tracking



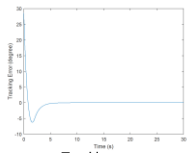
Control input u



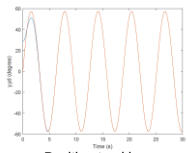
Control input τ

§6. Sliding Mode Control Based on Low Pass Filter

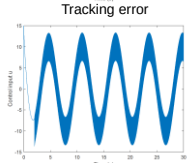
Results with low pass filter $\lambda = 25, \lambda_1 = 30, \lambda_2 = 30, \eta = 80$



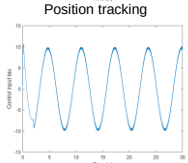
Tracking error



Position tracking



Control input u



Control input τ